# Sahaj Bhadja

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## Summary

Software Engineer with 2+ years building **education technology** and full-stack web apps with **TypeScript**, **React**, Next.js, Java Spring Boot, and Python on AWS, using AI-powered learning features, SQL, automated tests, and code review.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Advanced NLP, Analysis of Algorithms

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Data Retrieval, Operating Systems

## Experience

**USC Dornsife** **September 2025 – Present**
*Software Engineer* *Los Angeles, CA*

- Built backend and data infrastructure for an **AI education platform** serving 500+ users, with row-level security protecting learner data and tuned SQL queries cutting dashboard latency from 1.2s to 350ms.
- Restructured fault-tolerant **LLM** job orchestration with FastAPI workers and failure isolation, processing 1K+ AI jobs at 3x throughput, and shipped integrations through CI/CD at 99%+ reliability across 5K+ monthly jobs.

**Grad8** **June 2025 – August 2025**
*Software Engineer Intern* *Los Angeles, CA*

- Built an **AI-powered grading platform** with a React frontend and REST APIs for question upload, rubric generation, grading, and grading history, serving 150+ instructors and 10K+ daily requests.
- Integrated GPT-4o, Claude, and Qwen-VL to give students **feedback with rationale**, diagnosing OCR timeouts from logs and adding fallbacks, reaching 90%+ consistency and cutting feedback turnaround 60%.

**Aasma Technology Solutions** **August 2023 – July 2024**
*Software Developer Engineer – SecureSync* *India*

- Worked in a **shared codebase** with Microsoft and GitHub stakeholders in Agile sprints, building object-oriented Java and Python modules with 20+ **automated tests** run nightly in Jenkins.
- Reproduced and fixed slow API integrations by tracing failure logs, then added pagination, field filtering, and caching, improving response times 30% and cutting technical debt 20% for low-bandwidth users.

**Aasma Technology Solutions** **May 2023 – August 2023**
*Full Stack Developer Intern – VibrantChem* *India*

- Built **TypeScript React** interfaces and Java Spring Boot REST APIs for records, role-based access, and notifications, syncing 2M+ records across locations with real-time WebSocket updates for every site.
- Took part in **code review** and rollout across 6+ workflows, gating releases on Jest and Selenium tests in CI/CD, and used analytics on drop-off points to guide fixes that cut wastage 42%.

**LST Capitals** **November 2022 – January 2023**
*Web Developer Intern* *India*

- Built a React and TypeScript web dashboard with Flask APIs and Firebase real-time status updates, keeping core responses under 200ms and improving **cross-platform** compatibility 15% across devices.
- Designed system flows in Figma and **presented the architecture** to the client, then automated refund and dispute handling through idempotent webhooks, cutting manual payment work about 70% overall.

## Skills

**Languages:** TypeScript, JavaScript, Java, Python, SQL, HTML, CSS
**Frontend:** React, Next.js, TailwindCSS, Responsive Design, WebSockets, Figma
**Backend:** Spring Boot, FastAPI, Flask, Node.js, REST APIs, Microservices, Docker, Git, Agile, Linux
**Data & AI:** PostgreSQL, MySQL, Azure SQL, LLM APIs, Prompt Engineering, RAG, Embeddings
**Quality & Cloud:** Jest, Selenium, PyTest, Mockito, Unit Testing, Code Reviews, Debugging, CI/CD, GitHub Actions, AWS

## Projects

**AI Dialogue Practice Coach** | *Next.js, TypeScript, LLM APIs, Spring Boot, PostgreSQL*

- Designed a **conversation practice** app where students rehearse disagreements with an AI partner holding a different view, then get feedback on listening and curiosity against a **rubric**.

**Accessible Course Progress Dashboard** | *React, TypeScript, Java, Spring Boot, PostgreSQL*

- Designed an instructor dashboard showing learner progress, activity completion, and cohort trends, built to **WCAG** standards with keyboard navigation, screen-reader labels, and **automated tests** covering each view.

**Big-Brain** Live | *LLaMA-2, FAISS, Pinecone, Clerk, Convex, Shadcn*

- Built an AI **document** platform for document Q&A, summarization, and real-time collaborative conversations over 50K+ documents, with secure Clerk sign-in, cutting document review time 60% for 200+ users.